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FACTORING AND WEIGHTING APPROACHES TO STATUS SCORES AND CLIQUE IDENTIFICATION

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INTRODUCTION

Three very different techniques for devising indices of popularity or centrality have, with small variation, one basic solution for symmetric social structures. The resulting popularity or centrality scores have the advantage that they can be interpreted in all three different ways, and the isolation of the essential identity of the techniques makes more visible the important small differences between them. Moreover, it will be shown that this solution gives the clique structure at a glance. The solution will be contrasted with Hubbell's (1965) method of calculating status scores.

In the consideration of the three approaches, W will be a symmetric matrix of relationships or sociometric choices. All values in W are assumed to be between 0 and 1 inclusive. For example, it might be that $W = 1$ if persons i and j are friends and $W_{ij} = 0$ otherwise. The elements on the main diagonal are all zero.

The solution described in this paper only works for symmetric structures. Some social structures are naturally symmetric or can reasonably be made so. For example, talking to one another, spending time together, dating, and other behavioral relations are naturally symmetric. Friendship could be defined to include only reciprocal choices. An index of centrality in a communication structure might be desired, as a clue to how fast information will spread from an individual or to how much power an individual has because of his key position in the communication network. This structure will be symmetric if the communication channels are two-way. However, other relations will be inalterably asymmetric, such as nominations for the most powerful or popular in the group, and for these the method is not the appropriate tool. By being restricted to symmetric structures the technique is less general in its application but more general in its interpretation.

THE FACTOR ANALYSIS APPROACH

The bonds that form in a group could be conceived of as the result of 'interaction potentials' possessed by each individual. If W were a matrix of friendships, S_i would be individual i 's propensity to form friendships. The actual bond between i and j , W_{ij} , should be close to $S_i S_j$. In matrix notation, we want to calculate a column vector S

such that the sum of the squared differences between SS' and W is minimized. W is like a peculiar correlation matrix with zero communalities and the criterion for S is identical to the criterion for the first principal components factor of a correlation matrix (i.e., minimizing squared differences). The vector S (and the first principal components factor) is the eigenvector of the largest eigenvalue of the matrix standardized so that its length is the eigenvalue.

An interesting byproduct of the factor analysis is that there is an exact parallel between the factor structure and the clique structure. Define a clique as a set of individuals such that no relationships extend outside the clique and all individuals within the clique are related directly or indirectly through chains of choices. Each clique will be represented in the factor structure by a factor all of whose elements are greater than or equal to zero (Appendix C). The members of the clique will have nonzero positive loadings on this factor and all those who are not members of that clique will have zero loadings (Appendix A). Thus, the clique structure can be read at a glance; each clique will have its own vector of popularity scores.

The individuals who load high on a factor all of whose elements are greater than or equal to zero will be especially popular in their clique. For each clique the magnitude of the eigenvalue of its eigenvector of popularity scores will be a measure of how good the eigenvector is in summarizing the relationships in the clique, just as in a factor analysis of a correlation matrix these eigenvalues tell how much of the variance each factor accounts for.

It is well known that W is factorable even when it is not symmetric. If W is asymmetric it is possible to factor it into a row and a column factor whose product is a least squared error estimate of W (Wright and Evitts, 1961). The row factor is an eigenvector of WW' and the column factor is an eigenvector of $W'W$. However, the three approaches are the same when W is symmetric; if W were not symmetric the weighting techniques to be described next need not have any solution at all.

THE CONVERGENCE OF AN INFINITE SEQUENCE

A simple measure of the popularity is just the number of friends each person has. It might be desired to create second order indices of popularity by weighting each person's choice of others by the number of choices he receives. But there is no reason to stop at this point. If the second order indices are better than the first order indices (the number of friends), then third order indices in which each person's choice of others is weighted by his second order measure of popularity should be even better.

Let S_0 be a column vector of ones. Then S_1 , the first order popularity measure, is just $S_1 = WS_0$. The second order measure is $S_2 = WS_1 = W^2S_0$. The m th order measure is $S_m = W^mS_0$.

These popularity measures tend to become infinitely large. However, there is a small modification that does allow S_m to converge to a set of nonzero popularity scores as m approaches infinity. Let λ_1 be the largest eigenvalue of W . At each stage divide that result by the largest eigenvalue. $S_m = WS_{m-1}/\lambda_1 = W^mS_0/\lambda_1^m$. S_m converges to an eigenvector of λ_1 (Appendix B). Thus, the solution is almost identical to the factor analytic solution. The final popularity scores are described by an eigenvector

of the largest eigenvalue. This is equivalent to the most powerful factor in a factor analysis, if the eigenvector is standardized so that its length is the eigenvalue. However, if there is more than one clique the limit of the S_m scores will *only* give the popularity scores for the clique with the largest eigenvalue, which will tend to be the largest clique; entries for all nonmembers of the clique, including both isolates and those who belong to smaller cliques, will be zero. Thus, this approach is a little less general than the other two.

SIMULTANEOUS LINEAR EQUATIONS

Suppose we look at the limit of the previously described process. We want to weight each person's contribution to the popularity of others by his own popularity. We can set up a system of homogeneous linear equations for the unknown popularity scores. For each individual i :

$$S_i = W_{i1}S_1 + W_{i2}S_2 + \dots + W_{in}S_n$$

Each popularity score should be positive or zero.

This system of equations in matrix form is $S = WS$, or $(W-I)S = 0$, where the vector S is the unknown popularity scores. These equations have a nonzero solution only under the unlikely condition that $\det(W-I) = 0$. We could modify the equations above by multiplying the left hand side by a constant λ . This modification does not violate the spirit of the model and it allows a solution to the equations.

$$\lambda S_i = W_{i1}S_1 + \dots + W_{in}S_n$$

Now the equations in matrix form are $WS = \lambda S$, or $(W-\lambda I)S = 0$. This is the familiar problem of finding eigenvalues and eigenvectors. λ is an eigenvalue and S is an eigenvector.

Although every W will have eigenvectors and eigenvalues, the question remains as to whether there is a desirable solution. It was intended that λ should be positive and that each S_i should be positive. There should be just one solution; we should not be faced with an arbitrary choice among conflicting eigenvectors. In Appendix C it is demonstrated that each clique will have a positive λ such that all elements of its eigenvector are greater than or equal to zero. Moreover, Appendix A shows that these solutions do not contradict one another.

Thus, all three approaches have basically the same solution. The solution is always the eigenvector of the largest eigenvalue for each clique. There are minor differences. In the factor analysis approach the eigenvectors are standardized so that their lengths are their eigenvalues. The limiting process gives a solution only for the clique with the largest eigenvalue, which makes no difference if there is just one clique.

HUBBELL

The closest approach to that presented here is Katz's (1953) method of calculating status indices and Hubbell's (1965) method of clique detection, which also produces

status indices. Since Hubbell's technique includes Katz's, only Hubbell's will be discussed here.

Hubbell's basic equation for status scores is (1965:382):

$$S = E + WS$$

S is a column vector of status scores. W is called by Hubbell a 'structure' matrix. E is a vector of 'exogenous contributions' to status, aspects of a person's status that are not reflected in his nominations by other group members. Each individual's status is the sum of his nominations by others weighted by *their* status plus contributions of other factors (represented by e_i).

There is no discussion in Hubbell's article of how the e_i 's are to be determined. It is suggested that in the absence of external evidence E might as well be a vector of 1's. Thus, it seems fair to say that with sociometric data, in which W is given but not E , E is a mathematical convenience that turns a set of unsolvable homogeneous equations ($(W-I)S = 0$) into a set of solvable nonhomogeneous equations ($(W-I)S = E$), just as λ was introduced as a convenience.

It is easy to see that Hubbell's technique is similar to the technique proposed in this paper. What are their relative advantages and disadvantages?

1. Hubbell's approach requires the *arbitrary* assumption of some E vector. This appears to be true of the technique when it is applied to sociometric choice data rather than to the economic problems it was designed for. On the other hand, λ in the approach described in this paper is not arbitrarily supplied by the researcher; it comes from the data itself.
2. The solution vector S in the 'factoring-weighting' model of this paper has a data reduction interpretation. SS' is a least squared error estimate of W . This is not an interpretation that fits Hubbell's model.
3. Cliques can easily be identified using the factor-weighting approach. Hubbell's method produces a refined measure of the relationship between every pair of group members, which is the sum of all the direct and indirect paths between them, but cliques must still be grouped together 'by hand.'
4. Hubbell's technique, on the other hand, has a nice interpretation that the factor-weighting technique lacks. The elements of S are the sums of all the direct and indirect paths between a given individual and all other individuals if S is calculated using Hubbell's technique.
5. Hubbell's approach permits negative values in the matrix W and, most importantly, allows W to be asymmetric.

CONCLUSIONS

Three different approaches to calculating popularity scores have almost the same solution in symmetric structures. This is an economy; three approaches are reduced to just one. This is the main point of the paper. At the same time, some small but important differences are noted between the approaches. There is an infinity of 'weighting' solutions; only the normalized vectors are 'factor' solutions. The procedure based on weighting each choice by the number of choices received by the chooser

gives a solution only for the largest clique and makes members of smaller cliques appear to be isolates.

As a bonus, the factor solution and one of the weighting solutions give a clique structure when the matrix W is decomposable. The eigenvector for each submatrix in the decomposition is the vector of popularity scores for a clique.

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APPENDIXES

Appendix A

In Appendix C it is shown that there is associated with each clique an eigenvector all of whose elements are greater than or equal to zero. Here we wish to show that if an eigenvector of the matrix W has all positive elements the relationship between any of the individuals with strictly positive entries and any of the individuals with zero entries will be zero, and thus the eigenvectors with all positive or zero elements separate the system into sets of individuals who are not related to each other.

1. We have the following system of equations: $(W - \lambda I)S = 0$, where $W_{ii} = 0$, $W_{ij} \geq 0$ for every i and j . $\neq$
2. Suppose that a solution exists for which $S_1 > 0$, $S_2 = S_3 = \dots = S_n = 0$. An examination of the above equation shows that this implies that $W_{21} = W_{31} = \dots = W_{n1} = 0$. The first group does not overlap with any other group.
3. Suppose that a solution exists for which $S_1 > 0$, $S_2 > 0$, $\dots$, $S_k > 0$, $S_{k+1} = S_{k+2} = \dots = S_n = 0$, where $k \geq 2$.

The last $n-k$ equations now express $n-k$ relationships among the first k (strictly positive) status scores. All the relationships sum to zero and the $k(n-k)$ coefficients are all the relationships between the first k and the last $n-k$ individuals. If any of the relationships were greater than zero others would have to be negative because every sum is zero and the first k status scores are strictly positive. But no relationships are negative. Therefore all of these relationships W_{ij} , $i > k$, $j \leq k$, are zero.

Appendix B

We wish to show that a sequence of status scores converges to an eigenvector of the largest eigenvalue of W .

Let λ_1 be the eigenvalue of W that is greatest in absolute magnitude, assume it is of multiplicity one, and that it is strictly greater in absolute value than any other eigenvalue. The qualifications will almost certainly be true for any actual matrix W . Let $\{u_1, u_2, \dots, u_n\}$ be a set of orthonormal eigenvectors for W , where u_1 is the eigenvector of λ_1 . The eigenvalues of W are real and the eigenvectors are orthogonal because W is symmetric. It is shown in Appendix C that λ_1 is positive and that every element of u_1 is greater than or equal to zero (or else every element is less than or equal to zero—it makes no difference).

Let S_0 be a column vector of ones. Because the orthonormal eigenvectors of W are a basis for the vector space, $S_0 = \sum C_i u_i$. Multiplying through by u_1' , $C_1 =$

$u_1'S_0 > 0$, because every element of u_1 is greater than or equal to zero but at least one element is nonzero. Define $S_m = WS_{m-1}/\lambda_1$.

$$\begin{aligned} S_m &= W^m S_0 / \lambda_1^m = W^m \sum C_i u_i / \lambda_1^m = \sum C_i (\lambda_i / \lambda_1)^m u_i \\ &= C_1 u_1 + \sum_{i \geq 2} C_i (\lambda_i / \lambda_1)^m u_i \end{aligned}$$

The last term approaches zero as m increases. Therefore, S_m approaches $C_1 u_1$, which is also an eigenvector of λ_1 because $C_1 \neq 0$.

If the division were by any number smaller than λ_1 in absolute value S_m would diverge. If division were by any greater number S_m would converge to a zero vector. If division were by a number of the same magnitude but of opposite sign, S_m would oscillate.

Appendix C

We want to show that for each clique there exists a positive eigenvalue with an eigenvector all of whose elements are greater than or equal to zero. However, the eigenvalues and the eigenvectors of the matrix W are just the eigenvalues and the eigenvectors of the submatrices associated with each clique. Therefore, all that will be shown is that the eigenvalue of W that is largest in absolute magnitude is also positive and that it has an eigenvector all of whose elements are greater than or equal to zero. Then it will follow that an eigenvector of the largest (positive) eigenvalue of the submatrix associated with each clique has elements all of which are greater than or equal to zero.

1. Let W be a symmetric matrix all of whose elements are greater than or equal to zero. Let $\{\lambda_1, \lambda_2, \dots, \lambda_n\}$ be its eigenvalues and let $\{u_1, u_2, \dots, u_n\}$ be a corresponding set of orthonormal eigenvectors. The eigenvalues of W are real and the eigenvectors are orthogonal because W is symmetric. Assume, as in Appendix B, that λ_1 is greater in absolute value than any other eigenvalue.

$$u_1 = \begin{bmatrix} u_{11} \\ u_{21} \\ \dots \\ \dots \\ u_{n1} \end{bmatrix} \quad \text{Assume without loss of generality that } u_{11} \neq 0.$$

3. Let $x = \begin{bmatrix} 1 \\ 0 \\ \cdot \\ \cdot \\ 0 \end{bmatrix}$

Because the eigenvectors of the symmetric matrix W span the vector space $\sum c_i u_i = x$. Multiplying through by u_1' , $c_1 = u_1'x = u_{11} \neq 0$.

4. Define x_m as the first column of the m th power of the matrix W . Because every element of W is positive or zero, it is clear that every element of W^m is also positive or zero, and hence every element of x_m is positive or zero.
5. $x_m = W^m x = W^m \sum c_i u_i = \sum c_i W^m u_i = \sum c_i \lambda_i^m u_i$

$$= c_1 \lambda_1^m u_1 + \sum_{i \geq 2} c_i \lambda_i^m u_i = u_{11} \lambda_1^m u_1 + \sum_{i \geq 2} c_i \lambda_i^m u_i$$

6. We know that $\lambda_1 \neq 0$, $u_{11} \neq 0$. Therefore,

$$x_m / (u_{11} \lambda_1^m) = u_1 + (1/u_{11}) \sum_{i \geq 2} c_i (\lambda_i / \lambda_1)^m u_i$$

As m approaches infinity, the last term approaches zero because λ_1 is greater in absolute value than any other eigenvalue. Therefore, $x_m / (u_{11} \lambda_1^m)$ approaches u_1 . All elements of x_m are positive or zero. Therefore, all elements of u_1 are greater than or equal to zero or they are less than or equal to zero, depending on the sign of $u_{11} \lambda_1^m$. If u_1 is an eigenvector, so is $-u_1$. Thus λ_1 has an eigenvector with all elements greater than or equal to zero.

7. Moreover, $\lambda_1 > 0$, because if λ_1 were negative, $x_m / (u_{11} \lambda_1^m)$ would oscillate from positive to negative values, instead of approaching the nonzero vector u_1 .